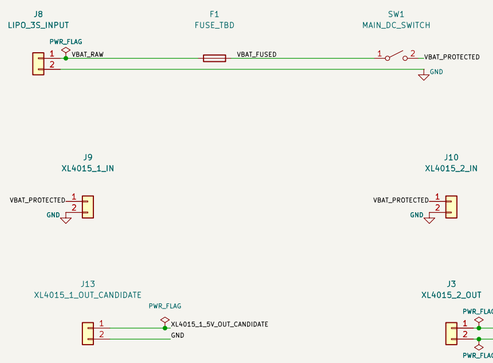
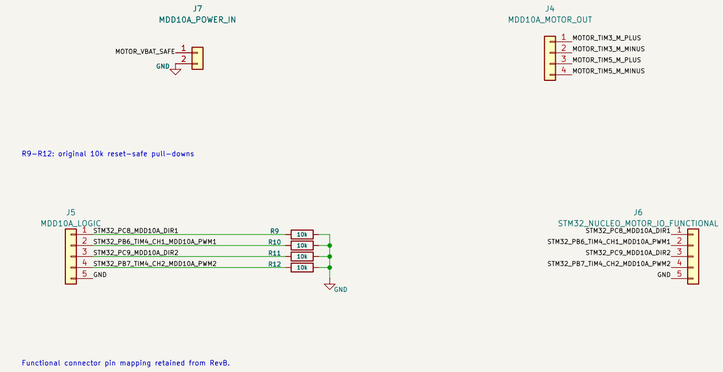
TRACKED MOBILE ROBOT | MVP WIRING | RevB-WIP

Graphical reorganization: 2026-11-06 | Original electrical checkpoint: 2026-08-12 | Net names link functional blocks

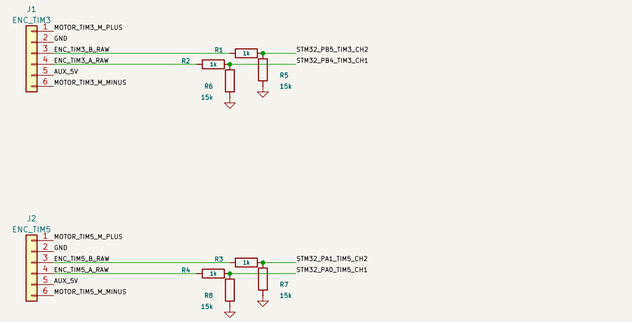
01 BATTERY & DC/DC INTERFACES



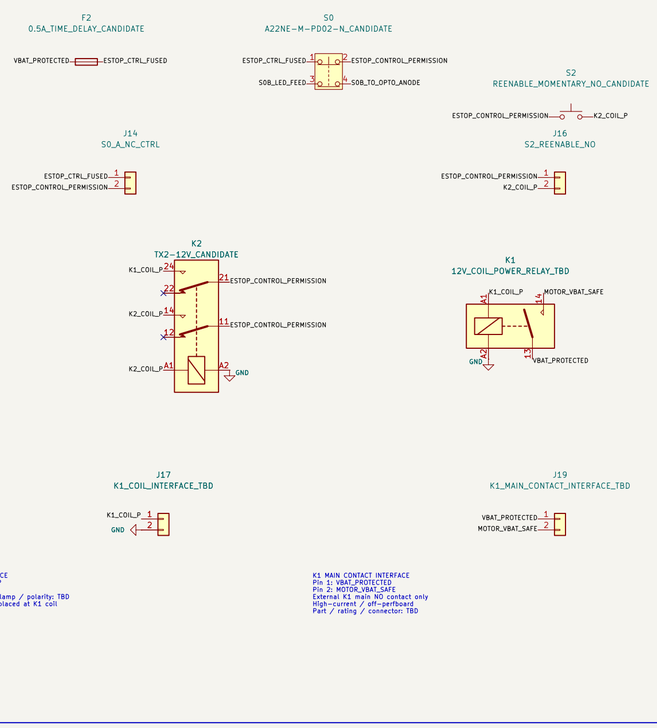
02 MDD10A & STM32 MOTOR I/O



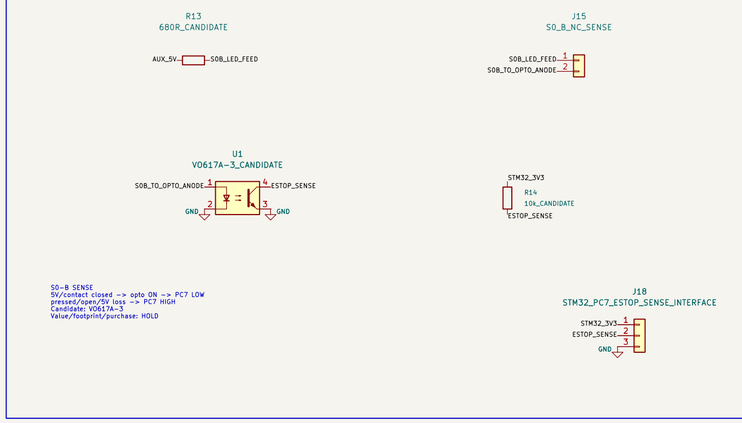
03 DUAL ENCODER CONDITIONING



04 E-STOP CONTROL / RELAY INTERFACES



05 E-STOP SENSE / STM32 PC7



06 STM32 – ESP32 UART



07 TEST POINTS



08 ORIGINAL CHECKPOINT STATUS

HISTORICAL: the status text below belongs to the original RevB-WIP checkpoint.

DESIGN STATUS

BENCH-VALIDATED
- Battery -> fuse -> switch power path
- MDD10A PWM/DIR static routing
- MDD10A DIR/PWM 10k pull-downs; reset capture PASS
- Dual encoder conditioning and hand-count
- STM32 <-> ESP32 UART, 115200 8N1

TBD / CANDIDATE
- RevB permanent wiring continuity
- Board power / USB backfeed policy
- Physical E-stop K1/K2/S0/S2 implementation
- Final F1/K1/main-wire rating
- Vehicle left/right powered channel assignment
- BNO085 power and I2C wiring

Layout review does not establish current as-built wiring or a new bench PASS.
Later hardware changes must be reconciled before using this as an as-built drawing.

09 RELAY MODEL & DRAWING NOTES

K2 CONTROL RELAY
Candidate: Panasonic TX2-12V
A1/A2: coil
11-14: HOLD NO
21-24: K1 ENABLE NO
12/22: NC unused
Footprint / purchase: HOLD

48 physical components / original references and connections retained.
Candidate and TBD markings are intentionally preserved.
No change to firmware, PCB layout, component values, or physical hardware.

LAYOUT REVIEW 2026-11-06
Functional blocks / label spacing / field alignment
Verification artifacts: assets/logs/pcb_design/
2026-11-06_mvp_schematic_layout_review

MDD10A reset-safe 10 kOhm pull-downs integrated: E-stop/power TBD
Status: WIP – Pull-down checkpoint

Sheet: /	
File: Tracked_Mobile_Robot_Wiring_RevB.kicad_sch	
Title: Tracked Mobile Robot Wiring	
Size: A2	Date: 2026-08-12
KiCad E.D.A. 10.0.5	**Rev: RevB-WIP**
	Id: 1/1